



Road Running

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Performance Demands of Road Running

This chapter describes the sport of road running with a particular focus on the marathon. Various factors contributing to elite distance runner success will be discussed, in parallel with the application of high-intensity interval training (HIIT).

Sport Description and Factors of Winning

Road running involves a race run over a measured course on an established road as opposed to an off-road cross-country event or athletics track (i.e., middle distance, chapter 13). The sport's popularity is highlighted in a 2015 statistic, in which there were more than 17 million reported finishers from more than 30,000 events held across all road running distances. Road running events are classified as long-distance in athletics terminology and range from 10 km to 42.195 km (the marathon; 26.2 miles). World-class times to complete such events for men and women range from 26 to 27 min and 30 to 31 min for the 10 km, to 2:03:00 to 2:08:00 and 2:15:00 to 2:20:00 for the marathon, respectively (30). A range of factors, including the topography and layout of the course, environmental conditions, tactics, and strat-

egy, influence success in each of these events (37). However, an athlete's physical capacity is fundamental to successful performance. Figure 14.1 shows the relative importance of road running skills (mechanics), tactics, and physical capacities, alongside other sports covered in the book.

The main physiological factors of importance for distance running success include a high maximal aerobic power (7, 8, 31, 53), the ability to perform at a high fractional utilization of maximal aerobic power throughout the duration of a race (31, 61), and running economically, or possessing a low energetic demand at a given running velocity (3, 31, 61). Key factors that differentiate top- versus high-class elite marathon runners include a high peak oxygen uptake and the capacity to draw on a fast running speed (i.e., > marathon pace) in a prefatigued state (8). Other factors associated with exceptional running performance include the ectomorphic somatotype, consistency of aerobic training from a young age, and a high motivation to succeed in the sport (61). A synopsis of these factors is shown in figure 14.2.

Relative Contribution of Physical Performance

Road running, particularly the marathon, differs from middle- to long-distance track running or



Figure 14.1 The position of the road runner on the three axes illustrates the relative importance of the three main physical capacities of importance for elite participation, acknowledging this varies slightly between race distances. Outside of critical psychological aspects, which are difficult to quantify, the pie chart shows the general relative importance skill, tactics, and physical capacities for road running success.

Adapted from G.A. Nader, "Concurrent Strength and Endurance Training: From Molecules to Man," *Medicine & Science in Sports & Exercise* 38, no. 11 (2006): 1965-1970.

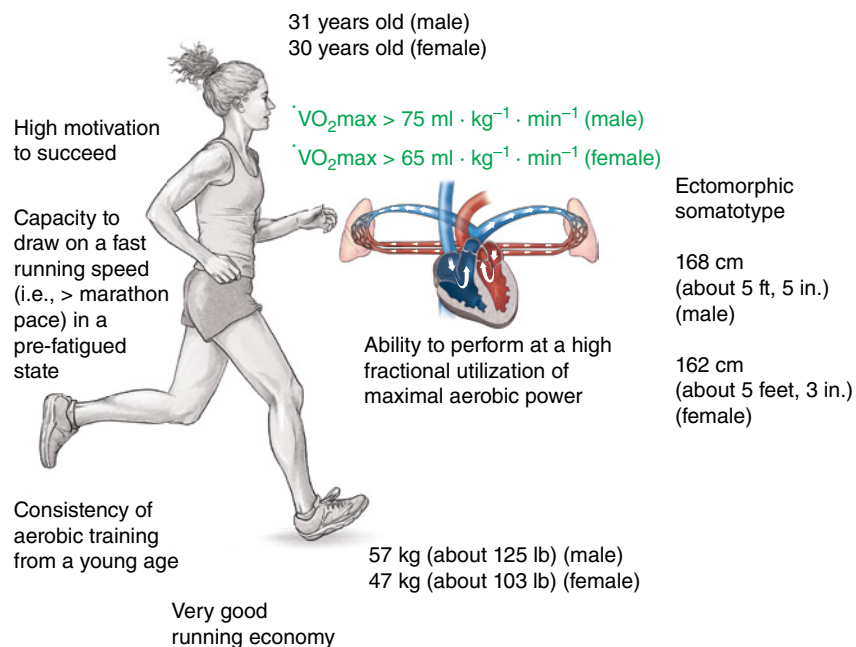


Figure 14.2 Key traits and factors identified as being related to successful marathon performance in elite-level road runners.

cross-country running with respect to the fact that tactical changes in pace are less defining to the overall performance outcome (40, 48). The distance of the marathon, coupled with large incentives for athletes to run the fastest time possible rather than simply being first across the finish line, means that a marathon runner can know the precise physiological requirements of his next race. For example, if an event is likely to be won in a time of ~2:05:00, or if there is monetary incentive (30) to break an existing course record of 2:04:30, then the required pace and associated physiological parameters needed to achieve this time can be reverse engineered. HIIT sessions can then be prescribed and tailored appropriately to meet the athlete's current status alongside the physiological demands associated with the known running performance. There are instances where the race may not be as fast as expected, but this may be due to a lack of direct financial incentive or environmental factors such as hot conditions due to the timing of an event (i.e., World Championships and Olympic Games).

Targets of Physical Performance in Road Running

The key adaptation targets of HIIT for marathon athletes are based on the physiological factors that are critical to marathon performance as described in the previous section (i.e., aerobic power, high fractional utilization of aerobic power, and economy). Therefore, prescription of HIIT for marathon runners should be focused on maximizing adaptations that benefit the physiological factors critical to marathon performance.

Aerobic Power

As described in chapter 3, aerobic power, or the maximal volume of oxygen uptake ($\dot{V}O_{2\max}$), refers to the highest amount of oxygen that can be consumed during maximal exercise. When running at close to maximal velocity over distances >3000 m, oxidative metabolism contributes to >90% of the energy requirements (33). Therefore, the duration of the marathon dictates that aerobic metabolism must be the primary energy source (6). $\dot{V}O_{2\max}$ has been shown to differentiate elite marathon runners (8). Therefore, many believe that the accumulation of training time at $\dot{V}O_{2\max}$, or at velocities associated with $\dot{V}O_{2\max}$ ($v\dot{V}O_{2\max}$), is likely to be the most

effective training stimulus for improving $\dot{V}O_{2\max}$ (11). For training sessions specifically targeting this adaptation, total session time should allow athletes to accumulate ~10 min at $\dot{V}O_{2\max}$ (see chapter 5). In practice, this may be best achieved by minimizing the time between the end of the warm-up and commencement of the HIIT component of the session, and prescribing a work:rest ratio of >1 (e.g., 4 × 5 min with 2 min active recovery). Additionally, the velocity required to accumulate time at $\dot{V}O_{2\max}$ may be manipulated by performing HIIT on an incline or on flat terrain.

High Fractional Utilization of Aerobic Power

Logic dictates that it is impossible to maintain the same $\dot{V}O_2$ for a marathon as running maximally over 3000 m. Therefore, the ability to utilize the highest possible fraction of one's maximal capacity is another key trait required for successful marathon outcomes. Accumulation of time at or just above marathon velocity (75% to 80% $v\dot{V}O_{2\max}$ (6) up to >85% $v\dot{V}O_{2\max}$ in top-class marathon runners (8)) may be the most appropriate stimulus for improving the fractional utilization of aerobic capacity. This is supported by the fact that training at this intensity comprises 10% to 15% of an elite marathon runner's weekly training volume (e.g., 3 × 8 to 9 min at 90% $v\dot{V}O_{2\max}$ with 2 min active recovery, or 2 × 20 min at 85% $v\dot{V}O_{2\max}$ with 3 min active recovery at 70% $v\dot{V}O_{2\max}$) (7, 8). These longer duration intervals are thought to improve buffering capacity and/or metabolic aspects, including blood lactic shuttling and enhanced fat relative to carbohydrate oxidation rates (32).

Economy

Running economy can be improved by influencing metabolic, biomechanical, and/or neuromuscular efficiency (3). HIIT training, targeting $\dot{V}O_{2\max}$ and high fractional utilization of $\dot{V}O_{2\max}$, also has been shown to enhance biomechanical factors influencing running economy. For example, studies have shown improvements in running economy from 1% to 3% using HIIT completed on a 10% incline, such as 9 × 2 min at ~100% $v\dot{V}O_{2\max}$ with 4 min recovery (2), and 4 × 4 min at ~106% $v\dot{V}O_{2\max}$ with 2 min recovery (21). Neuromuscular efficiencies may also be improved by targeting changes in muscle/tendon characteristics through loading

patterns that exaggerate relevant mechanical needs of distance runners such as plyometric exercises. Within the specific context of HIIT, prescription of intervals completed downhill or on the decline (e.g., -10% gradient, 2×10 min intervals at 85% to 90% $\dot{V}O_{2\max}$ with 1 min recovery between sets (15)) can be utilized to target such adaptations.

Key Weapons, Manipulations, and Surveillance Tools

In this section, we present the different HIIT weapons, their manipulations, along with ways by which we can monitor their effectiveness (surveillance) for road runners. Like many endurance athlete programs presented throughout the book (i.e., rowers, cyclists, triathletes), the weekly training intensity distribution of an elite marathon runner generally follows the polarized model whereby about 80% of the training volume is completed at submaximal intensities, with the remaining 20% completed at marathon race pace or faster (7, 8, 50). The low- to medium-intensity (50% to 80% $HR_{\max}$, 40% to 80% $\dot{V}O_{2\max}$ i.e., <80% marathon race velocity) component of training provides an important foundation because it builds the resilience (psychological and neuromuscular/musculoskeletal) (33), body composition, economy (3), aerobic capacity (17, 18, 33), and fat oxidation rates (27) required for the marathon distance.

While HIIT typically forms only ~15% of the total training volume, it provides a stimulus that critically contributes to marathon performance (5-8). Fundamental HIIT targets for road running include types 1 through 4, with type 1 and 3 targets being particularly important for improving maximal aerobic power and fractional utilization of maximal aerobic power, and type 2 and 4 targets important for improving running economy.

HIIT Weapons

The overall breakdown of HIIT weapons used throughout a season in elite marathon runners is shown in figure 14.3. Given the requirements of the marathon, the dominant forms of HIIT tend to be long intervals, using work:rest ratios >1 and terrain (i.e., flat versus incline versus decline) manipulated to emphasize types 1 through 4 targets. Short-interval HIIT is also used to a much lesser extent to target types 2, 3, and 4 during specific phases of preparation (see figure 4.14) (12).

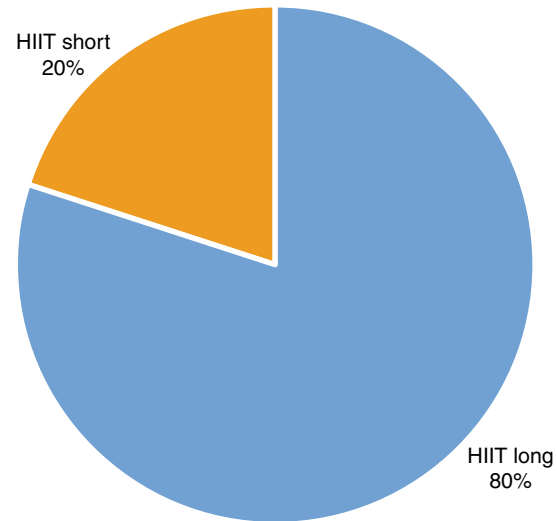


Figure 14.3 General percentage breakdown of HIIT formats (weapons) used in road running across a season at the elite level.

Manipulations of Interval Training Variables

The running intensity and modality of each HIIT session can be modulated such that a specific metabolic, neuromuscular, and locomotor response can be targeted. While manipulation of the work:rest ratio and volume are the primary means by which to modify metabolic HIIT targets, manipulation of the modality (i.e., uphill or downhill running) where HIIT is performed can target specific neuromuscular and locomotor responses.

HIIT With Long Intervals

Long-interval HIIT for road runners typically ranges from 60 s to 15 min work durations at intensities at or faster than marathon race velocity (85% to 100% $\dot{V}O_{2\max}$), with work:rest ratios >1 (see examples in figure 14.4). The total duration of time accumulated at these intensities is typically 20 to 40 min, which may be split into 1 to 5 sets of equal or varying duration. A classic long-interval HIIT session is 5×5 min at 90% to 95% $\dot{V}O_{2\max}$ with 2 min active recovery at 40% $\dot{V}O_{2\max}$. Two other common long-interval HIIT sessions for a marathon runner would be 25×400 m at 95 to 100% $\dot{V}O_{2\max}$ with 15 to 20 s recovery, or four sets of 5×400 m at 100 to 115% $\dot{V}O_{2\max}$ with 45 s recovery between repetitions and 60 s between sets (figure 14.4). The desired metabolic load for both sessions is similar. However,

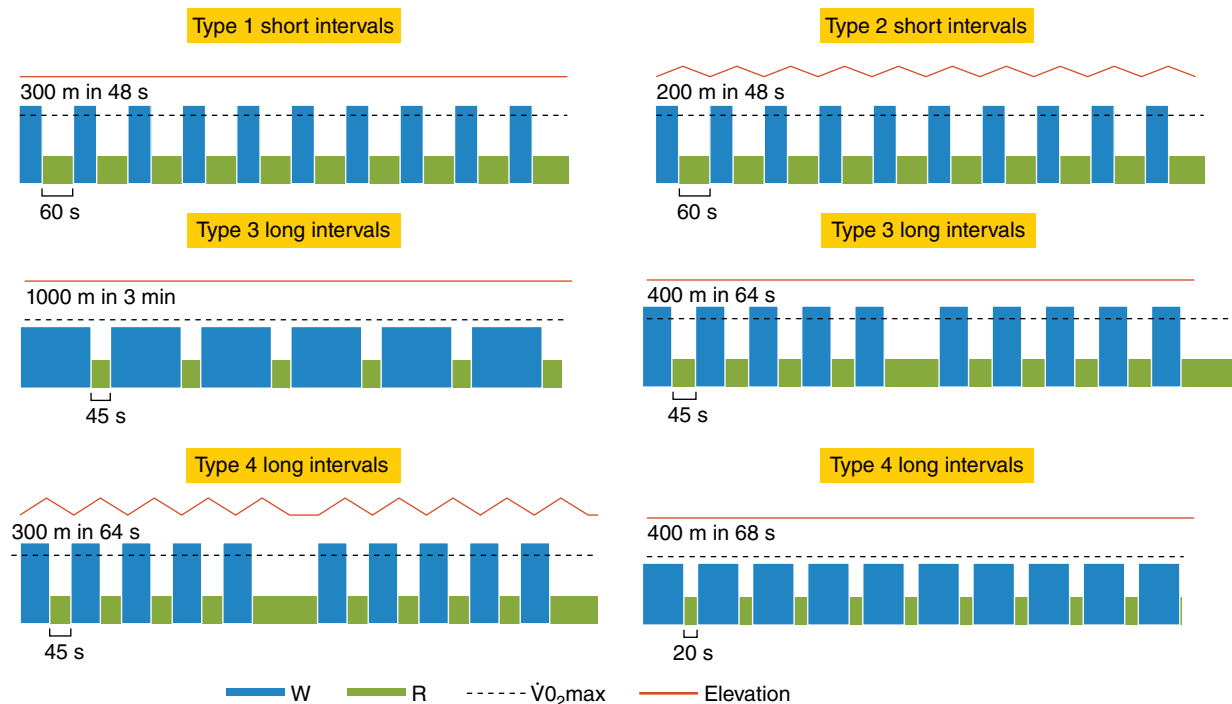


Figure 14.4 Examples of short and long intervals and the various HIIT target types commonly used in marathon training. Note the effect of manipulating work:rest ratios and modality (i.e., elevation, red line) on HIIT target type. W: work; R: recovery; $\dot{V}O_{2\max}$: level associated with maximal oxygen uptake.

the manipulation of the work:rest ratio results in different neuromuscular load. For example, the first session has shorter recovery and therefore a greater neuromuscular load (type 4), whereas the second session, broken into four sets with greater recovery duration and faster pace, has less neuromuscular load (type 3). Further, changing the modality and completing the second session on an incline (i.e., uphill) compared to on the flat, will increase the neuromuscular load, shifting the targets from type 3 to 4 (see figure 14.4).

HIIT With Short Intervals

Short-interval HIIT for road runners typically ranges from 30 s to 45 s work durations at intensities faster than marathon race velocity ($>110\%$ $\dot{V}O_{2\max}$), with work:rest ratios between 1:2 and 1:3. The total duration of time accumulated at these intensities is typically 10 to 20 min and can be done in combination with a long-interval HIIT session. An example session might be 3×2000 m at 90% $\dot{V}O_{2\max}$ with 90 s active recovery (type 3 long interval), 4 min of active recovery at 40% to 50% $\dot{V}O_{2\max}$, then 12×300 m at 115% to 130% $\dot{V}O_{2\max}$ with 60 s

active recovery (type 1 short interval). If the short-interval HIIT is done as a standalone session with passive recovery, the metabolic targets become type 4. However, combining it with the prior long-interval HIIT session allows all HIIT targets (types 1 to 4) to be met. Similar to long-interval HIIT, different HIIT targets can be achieved by manipulating the work:rest ratio and modality (see figure 14.4).

Load Surveillance and Monitoring Tools

Running is a bodyweight-loaded activity that involves both concentric and eccentric loading, therefore requiring large aerobic oxidative and neuromuscular demands. During high-volume training weeks that many elite road runners perform (~ 200 km), a runner may accumulate 148,000 to 168,000 steps (i.e., 740 to 840 steps per km, depending on speed). This means that for a typical marathon runner weighing 65 kg, and assuming the impact force of each step is ~ 3 times that of gravity (i.e., 195 kg per step), the total weekly load becomes 28 to 35 million kg (51). Therefore, load surveillance is of upmost

importance for ensuring that training load is progressed appropriately and the risk of lower-limb overuse injury is minimized.

With recent advances in technology, monitoring and/or quantifying individual session/workout and longitudinal training load has become highly data driven. Wrist-based global-positioning systems with built-in accelerometers and integrated heart rate (HR) monitors have made collection of key running metrics (e.g., instantaneous pace, stride frequency, and heart rate) accessible to the masses. This data can be used to provide insight into the physiological response to training and competition intensity and compared alongside laboratory test data to monitor changes in physiological capacities, allowing more specific prescription in training programs (e.g., prescribed pace to run an HIIT session involving type 1 versus 2 or type 3 versus 4 targets when the desire is to increase or decrease neuromuscular load). In addition, such data can be used to calculate training load. Most simply, training load can be defined as the product of exercise intensity and duration. Exercise intensity can be derived from either the external (distance, run pace, power output) or internal (oxygen consumption, heart rate, blood lactate concentration, rating of perceived exertion) loads (chapter 8). There are many models available to derive training load (19), including training stress score from power output (54), training impulse using heart rate (1), and session RPE based on the perceived exertion of the training session (20). Examination of longitudinal trends in training loads such as acute and chronic training load and training stress balance metrics are increasingly being used to model fitness/load progression to minimize injury risk (22, 23), with many commercial software packages now available with the necessary analysis tools to calculate these metrics (see chapter 8).

Training Status Surveillance and Monitoring Tools

There are many tools available to monitor an athlete's training status or readiness to train. The primary tool used by coaches who have regular (i.e., daily) contact with their athletes is to monitor training status or readiness to train using simple visual and audible feedback. This primarily involves a simple conversation with the athlete, asking, "How do you feel?" An experienced coach's intuition based on visual assessment of body language and attitude prior to or during a training session is also important. While

such parameters might be considered qualitative, this daily face-to-face contact is an invaluable surveillance tool used by the most experienced coaches. More quantifiable monitoring tools are generally limited to noninvasive measures such as daily wellness metrics (49), daily resting HR and heart rate variability (HRV) and exercise HR (10), and GPS/accelerometer-derived data. Specifically, analyzing trends in daily wellness, waking HR and HRV in relation to the training context (internal and external loads, i.e., exercise HR and GPS data) provide a powerful tool to confirm an athlete's readiness to perform or assess how an athlete is adapting to the training program (56) (see chapter 9). Although not as accessible to all athletes, blood testing can be used to inform on training status (e.g., overtraining state (34), adaptation to altitude (35), or heat stress (42)) in a similar way to how the biological passport system is used to monitor potential doping practices (44). A benefit of using quantifiable training status tools is that they provide historical information that can be retrospectively analyzed and used to plan future seasons or for understanding how an athlete has adapted to a specific training stimulus.

Strategies for Structuring the Training Program Using HIIT

Strategies for structuring HIIT programming in road running may be confined by various controllable and uncontrollable factors, before proper progression and periodization can be appropriately implemented. This section discusses some of the nuances of training program and HIIT structuring in road running. As discussed, marathon runners generally adopt a polarized training intensity distribution (50) throughout the entire season, with the specific contents of each HIIT session manipulated to suit the desired physiological outcome. Additionally, the timing of HIIT sessions within a weekly micro-cycle is often manipulated (e.g., from 24 to 72 h) to alter the physiological outcomes based on variable recovery kinetics (55).

Controllable and Uncontrollable Factors

Most elite marathon runners will focus on two or three competitions to peak for each year. Unless there are specific qualification requirements for particular events (e.g., world championships, Olympic Games), athletes have reasonable flexibility in choosing the

peak events they prefer to target because there are events all over the world each month of the year. This makes planning a season relatively simple compared to other sports that may have 40 to 90 competitions or race days per year (see chapter 15, Road Cycling).

Program Structure and Progression

Compared to many other sports, periodization for road running, particularly the marathon, is relatively straightforward. Most elite marathon runners typically target only two or three key events per year. The duration between these peak events (i.e., 12 to 25 wk) usually allows for progressive and periodized training blocks that can target key physiological adaptations (28). In general terms, the training phases of a marathon runner's season follow a classic progression (figure 14.5). As outlined in the previous

section on load surveillance, it is imperative that training load is progressed appropriately to minimize the risk of overuse injuries and illness.

Macrocycle

For most elite marathon runners, the season is periodized over a 12 month cycle with two or three primary competition objectives or peaks. Within each macrocycle, an athlete might complete 3 to 5 mesocycles progressing from general to specific to competition focus.

Mesocycle

Depending on the time between key events, mesocycles may vary from 3 to 6 wk, with 2 to 5 loading weeks separated by 1 or 2 recovery or de-loading weeks. Due to the weight-bearing nature of running, shock or block mesocycles of high training load

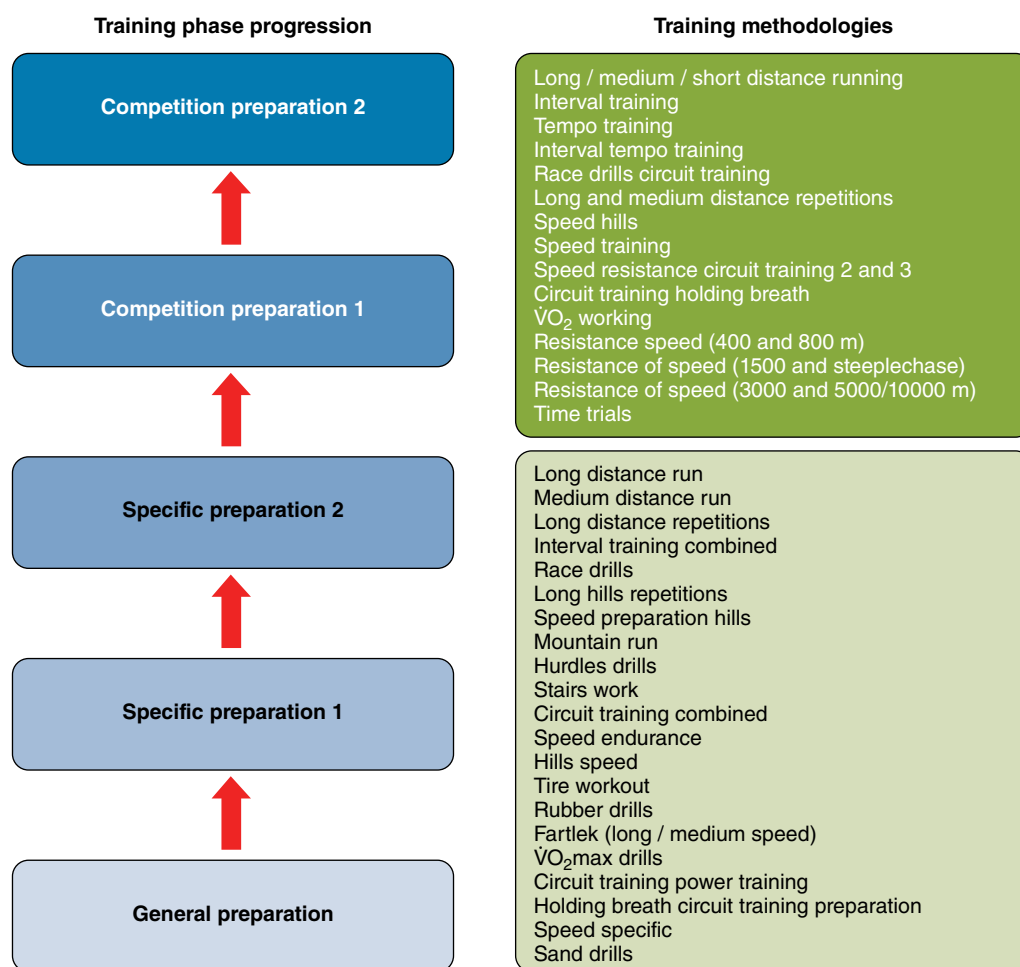


Figure 14.5 Progression of training phases (left panels) with the training methods (right panels) employed.

common in other endurance sports such as cycling (45, 46) or cross-country skiing (47) are typically not as used unless the additional load is composed of cross-training activities such as water running or cycling. This relates to the large neuromuscular demands of weight-bearing running as outlined.

Microcycle

The microcycle of a marathon runner is traditionally, but not always, structured around 7 days. During various phases of mesocycles, a microcycle may comprise up to 11 running-based sessions, 2 or 3 strength and conditioning sessions, 2 or 3 drill sessions, and 1 or 2 water-based sessions. The 11 running-based sessions may include 2 or 3 HIIT sessions (completed on the track, road, hills, or on a treadmill), 1 or 2 long-distance runs (>20 km), which may be steady or increasing in pace, as well as 4 to 6 medium-distance runs (10 to 15 km) at a steady moderate/easy pace. The timing and specific nature of weekly HIIT sessions varies depending on the mesocycle and the desired physiological adaptation objective. For example, during the early general preparation mesocycle, a 7 d microcycle may include 3 HIIT sessions separated by 1 or 2 d. Conversely, during a competitive season mesocycle, only 2 HIIT sessions may be prescribed separated by 2 or 3 d. Modulation of the timing between HIIT sessions may affect recovery kinetics, potentially resulting in differing adaptation responses (55). The detailed periodization examples across the different mesocycles in a world-class marathoner that follow offer real-world examples for the reader to consider.

Hypoxic Stress

While not all successful elite marathon runners undertake altitude training, there is general consensus in the athletic community and scientific literature that training at altitude *may* improve sea level performance (9, 25). The underlying physiological adaptations from training at altitude that may positively benefit performance include an increase in erythropoietin response increasing hemoglobin mass and oxygen-carrying capacity (26) and improved oxygen extraction and economy (25). These adaptations may enhance adaptations during a subsequent sea level training block or upcoming competition (9, 25), although not always (36). The minimal number of peak competitions a marathon runner competes in throughout a year means that periodization of altitude training blocks is relatively straightforward. However, the training content

(including the incorporation of HIIT sessions) in the lead up to and during altitude exposure can greatly influence the adaptation response, and individual athlete fatigue and recovery profiles need to be closely monitored (52). The HIIT response associated with hypoxic training is described in chapter 4.

Heat Stress

Many of the big city marathons are held during cooler times of the year. However, the Olympic Games and World Championship events are often completed in hot environments. Acclimating to the environmental conditions is common practice and generally involves training in a competition-specific (hot/humid) environment for 2 to 3 wk (43). For most athletes, it is not feasible to travel to such locations due to the associated costs involved. Therefore, alternative modalities can be used to provide the appropriate heat stress required to initiate the desired physiological adaptations, including passive sauna exposure (57) or training in an environmental chamber (24). Further, heat stress itself has also emerged as a potential ergogenic aid for enhancing various aspects of training adaptation (14) (see also chapter 4).

Periodized Nutrition

Given the high training volumes completed by marathon runners, nutrition naturally plays an important role in supporting both low-intensity training and HIIT sessions, as well as recovery and altering body composition. Additionally, manipulation of fuel availability (i.e., carbohydrate) before and after each training session can influence the physiological adaptation (see also chapter 4). For example, adopting a train high, sleep low periodization of carbohydrate availability approach, whereby an HIIT session may be performed with high carbohydrate availability but a subsequent low-intensity session the following day is performed with low carbohydrate availability, has been shown to improve submaximal economy, supramaximal capacity, and race pace performance (38). Additionally, the nutritional support required for HIIT or low-intensity training can affect the quality/intensity of the session, as well as the molecular pathways that drive adaptations within the muscle (16) (chapters 3 and 4). Therefore, incorporation of such strategies into the overall periodized plan needs to be considered (4, 29). On race day, the ability to effectively utilize carbohydrate and fat oxidation throughout the entire race will allow for the highest fractional utilization of aerobic

power. A recent study suggests that it may be the enhanced ability to use fat metabolism while exercising at high intensities that differentiates well-trained versus recreational endurance athletes (27), highlighting the importance of periodizing nutrition to promote enhanced fat metabolism (38). Nutrition on race day is also fundamental for achieving the full expression of performance capacity. Given the duration and intensity that the marathon is completed in, appropriate individualized nutrition is thought of as being vital for success (58). Adopting a low-residue diet during the days prior to competition may also help minimize potential gastrointestinal disturbances (13). The use of nutritional supplements such as dietary nitrates (e.g., beetroot juice) (39) and caffeine (41) have also been shown to improve economy and delay fatigue, however, athletes should trial any sort of supplement strategy well beforehand so as to be well familiarized prior to race day.

Incorporation of HIIT

As described before, HIIT can be incorporated into the training program during all phases of training. It is important that the overall adaptation response or goal be considered when planning a training program. This is because the timing between HIIT sessions within the week, the modality of HIIT sessions, and the interaction between varying stimuli such as other training, environmental stress, or nutritional strategies will influence the adaptation response.

Sample Training Programs

The following examples are from the specific 25 wk preparation of Brazilian marathon runner Carlos da Costa for the 1998 Berlin marathon, where he broke the world record at that time, crossing the finish line in a time of 02:06:05. This result was the first time in history that a man averaged <3 min per km for the



Brazilian marathon runner Carlos da Costa with coach Carlos Alberto Cavalheiro after setting the world record at the 1998 Berlin marathon in a time of 02:06:05.

Victor Sailer/Photorun.

marathon distance. As described in the previous section (figure 14.5), this preparation was broken into one 5 wk general preparation phase (figure 14.6), two 5 wk specific preparation phases (figure 14.7), and finally two 5 wk competition preparation phases (figure 14.8). In each mesocycle, the timing and nature of weekly HIIT throughout each microcycle is shown to change so as to target the desired adaptation response. In the general preparation phase (figure 14.6), there is an emphasis on HIIT target types 2, 3, and 4. Three HIIT sessions are programmed per weekly microcycle, with 48 to 72 h of recovery (low-intensity sessions) to allow more recovery. Neuromuscular load is also relatively high during this phase with the aim of developing resilience, strength, and economy. For example, two HIIT sessions per week include type 2 and 4 targets, supplemented with 2 weekly strength and conditioning (i.e., resistance training in the gym) sessions. During

the specific preparation phases, variations of long-interval HIIT type 3 predominate with the introduction of moderate intensity (race pace) training. There is reduced emphasis on neuromuscular loading (no HIIT targets but strength and conditioning sessions remain). Interestingly, the race pace session is programmed within 24 h of the second weekly HIIT session, which means the athlete is likely completing this session in a prefatigued or not completely recovered state. Finally, during the competition preparation phase (figure 14.8), all specific neuromuscular-loading sessions are removed (no HIIT types 2 or 4 or strength and conditioning). However, there is further emphasis on developing race pace ability with the incorporation of race pace work in the long-interval HIIT type 3 sessions. Due to the demanding nature (volume) of each HIIT or race pace session, at least 24 to 72 h separates them to allow for more complete recovery.

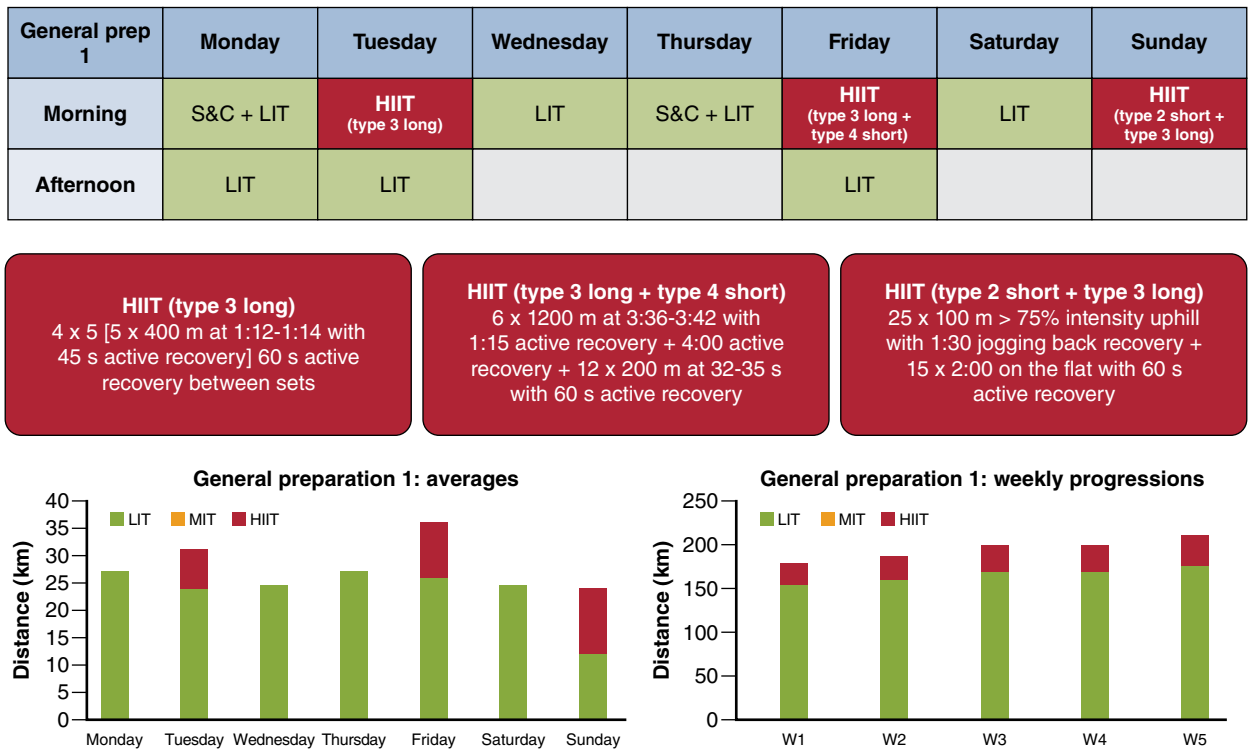


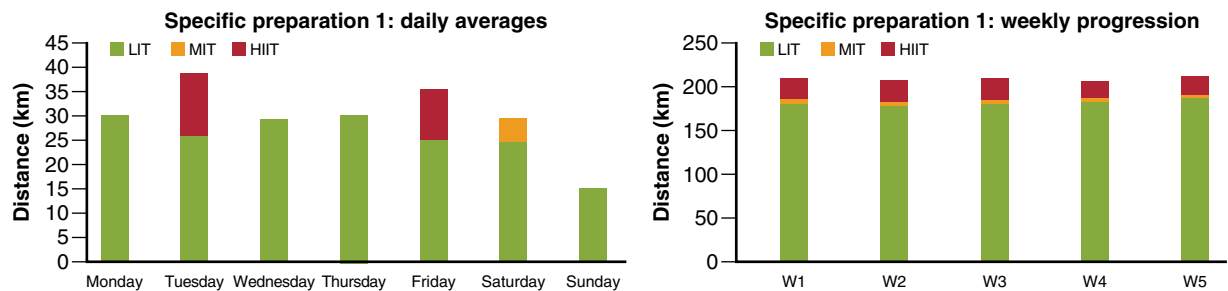
Figure 14.6 A sample general preparation mesocycle consisting of 5 microcycles (weeks). The basic structure and distribution of training sessions within each 7 d microcycle is summarized in terms of the strength and conditioning (S&C) training/resistance training in the gym; low-intensity training (LIT) at <75% $\dot{V}O_{2max}$, <75% HR_{max} ; moderate-intensity training (MIT) at 76% to 85% $\dot{V}O_{2max}$, 78% to 87% HR_{max} ; high-intensity interval training (HIIT) at >85% $\dot{V}O_{2max}$, >88% HR_{max} . Examples of specific HIIT sessions undertaken by former world record holder Carlos da Costa during the general preparation phase are provided. The average breakdown of intensity distribution throughout a microcycle and the weekly progression of volume and intensity distribution are described.

Specific prep 1	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Morning	S&C + LIT	HIIT (type 3 long)	LIT	S&C + LIT	HIIT (type 3 long)	LIT/MIT	LIT
Afternoon	LIT	LIT			LIT		

HIIT (type 3 long)
15 x 1000 m at 2:54-2:58 with 35-45 s active recovery

HIIT (type 3 long)
4 x [1600 m at 4:36-4:45 with 50 s active recovery, 800 m at 2:14-2:20 with 90 s active recovery]

LIT/MIT
25 x 100 m > 75% intensity uphill with 1:30 jogging back recovery + 15 x 2:00 on the flat with 60 s active recovery



Specific prep 2	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Morning	LIT	HIIT (type 3 long)	LIT	LIT	HIIT (type 3 long)	LIT/MIT	Rest or LIT
Afternoon	LIT	LIT			LIT		

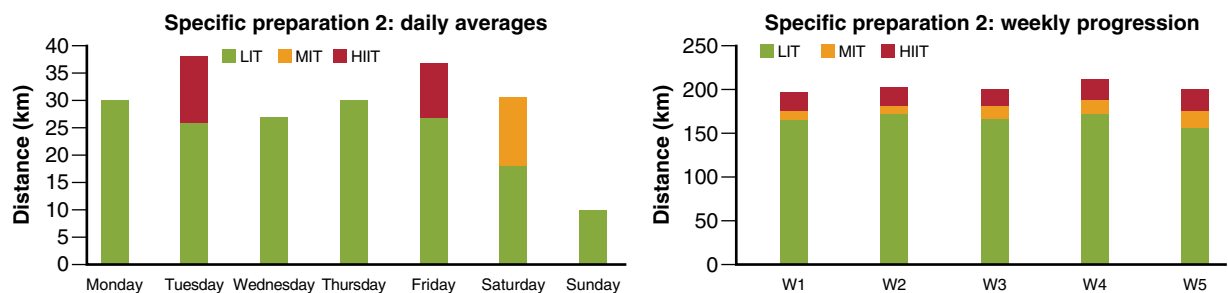


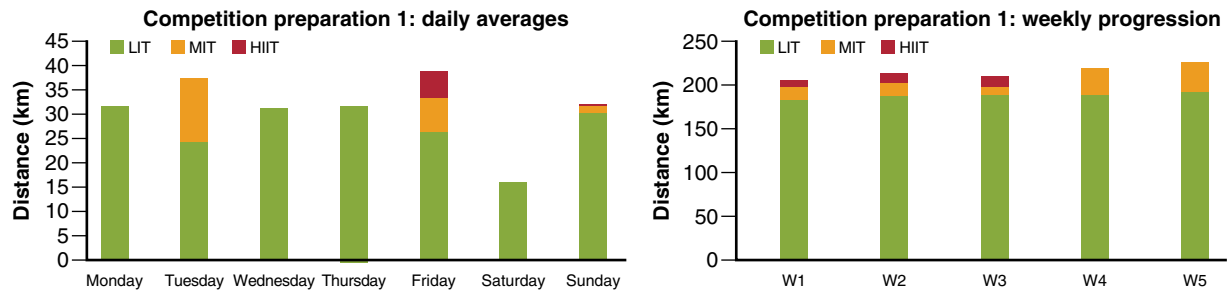
Figure 14.7 A sample of two specific preparation mesocycles consisting of 5 microcycles (weeks). The basic structure and distribution of training sessions within each 7 d microcycle is summarized in terms of the strength and conditioning (S&C) training/resistance training in the gym; low-intensity training (LIT) at <75% $\dot{V}O_{2\max}$, <75% $HR_{\max}$; moderate-intensity training (MIT) at 76 to 85% $\dot{V}O_{2\max}$, 78% to 87% $HR_{\max}$; high-intensity interval training (HIIT) at >85% $\dot{V}O_{2\max}$, >88% $HR_{\max}$. Examples of specific HIIT sessions undertaken by former world record holder Carlos da Costa during the general preparation phase are provided. The average breakdown of intensity distribution throughout a microcycle, and the weekly progression of volume and intensity distribution are described.

Competition prep 1	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Morning	LIT	MIT/HIIT (type 3 long)	LIT	LIT	HIIT (type 3 long)	LIT	LIT/MIT
Afternoon	LIT	LIT			LIT		

MIT/HIIT (type 3 long)
15 x 1000 m at 3:02-3:06 with
20-25 s active recovery

HIIT (type 3 long)
5 x [3000 m increasing pace for
each at 9:30-9:25-9:20-9:10-9:00
with 3:00 active recovery, 800 m at
2:14-2:20 with 90 s active recovery]

LIT/MIT
30 km long run progressing pace
every 10/8/5/4/2/1 km as 3:35-
3:30-3:25-3:20-3:15-3:10 over
flat terrain



Competition prep 1	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Morning	LIT	MIT/HIIT (type 3 long)	LIT	LIT	HIIT (type 3 long)	LIT	LIT/MIT
Afternoon	LIT	LIT			LIT		

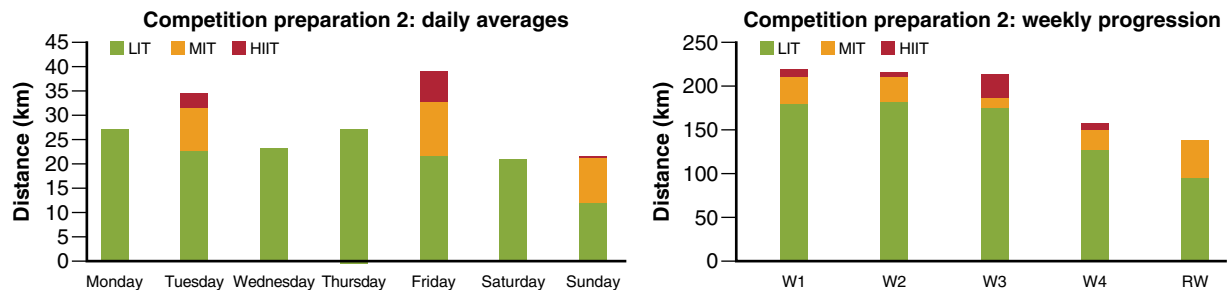


Figure 14.8 A sample of two competitive period mesocycles consisting of 5 microcycles (weeks). The basic structure and distribution of training sessions within each 7 d microcycle is summarized in terms of the S&C, strength and conditioning training/resistance training in the gym; LIT, low-intensity training at $<75\%$ $\dot{V}O_{2\max}$, $<75\%$ $HR_{\max}$; MIT, moderate-intensity training at 76 to 85% $\dot{V}O_{2\max}$, 78 to 87% $HR_{\max}$; HIIT, high-intensity interval training at $>85\%$ $\dot{V}O_{2\max}$, $>88\%$ $HR_{\max}$. Examples of specific HIIT sessions undertaken by former world record holder Carlos da Costa during the general preparation phase are provided. The average breakdown of intensity distribution throughout a microcycle, and the weekly progression of volume and intensity distribution are described.

CASE STUDY

Jamie Stanley

Often in elite sport, the perennial question on a coach's mind may be either "how can we do more?" or "how can we fit more into a program?" to further improve performance. While there is a strong relationship between the volume of training done and the performance outcome, this approach, if poorly implemented can lead to overtraining, maladaptation, injury/illness, and ultimately a compromised preparation and suboptimal performance outcome. As a sport scientist, I like to challenge this approach and ask the alternate question, "What is the minimum stimulus required to achieve the optimal adaptation and performance?" This is commonly termed today as the *minimum effective dose*. As a former elite-level long-distance triathlete accustomed to training 20 to 25 h per wk, I decided to adopt the minimum effective philosophy as a self-experiment during the personal preparation for my first marathon, with the following constraints:

goal time = <2 h 30 min

average training <80 km per wk

6 mo preparation time

About 14 mo separated my final elite half-ironman distance competition (1.9 km swim, 90 km cycle, 21.1 km run) of my career, before commencing my marathon training, during which I exercised only for social and health purposes (running, cycling, and resistance training), including a weekly training volume of 25 km per wk running, 184 km per wk cycling, and 80 min per wk resistance training.

I split my 6 month marathon preparation into three phases. The first phase, lasting 2 mo, was focused on building resilience to the increased running volume. During this phase, weekly run mileage progressed with predominately frequent low-intensity running, before being supplemented with one weekly specific downhill session (to improve neuromuscular and mechanical efficiency as discussed before). The second phase, lasting 3 mo, incorporated the bulk of the "work" and consisted of 6 to 8 runs per week. Each week included one HIIT session on flat terrain (rotation of Mona-fartlek, 90 s to 15 s short intervals at ~95% to 150% $\dot{V}O_2$ max; Yasso 800s, 5 to 10 $\times$ 800 m at ~115% to 130% $\dot{V}O_2$ max; and 3 or 4 $\times$ 2 km at ~95% $\dot{V}O_2$ max), one HIIT session or fartlek on hills (e.g., 4 $\times$ 4 min ~100% $\dot{V}O_2$ max) and the occasional 5 or 10 km road race. During these first two phases, weekly bike volume included ~60 km at low intensity to supplement the aerobic training load. Periodized weekly blocks of passive heat exposure (sauna and hot tub) and a 4 wk live-high, train-low hypoxic exposure (altitude tent) was implemented to further enhance cardiovascular and potential molecular adaptations.

The final phase included a 4 wk, three-phase taper whereby training volume and load were initially reduced to 40% to 50% of maximum for 1 wk, returning to 80% to 90% of maximum for the next 2 wk, before finally being reduced for race week. This nontraditional taper was inspired by my father (John Stanley, 02:17:04 marathon best, Australian national team during the 1970s and 1980s) and recent papers detailing preparations of elite Norwegian XC skiers (59) and orienteers (60).

Approximately every 4 wk across the entire preparation I completed a progressively longer long run, commencing with a 20 km steady state (~40% $\dot{V}O_2$ max [4:33 min/km]) progressing to the final run of 36.1 km (the first 15 km was run at ~65% $\dot{V}O_2$ max [3:59 min/km] and the final 21.1 km at marathon pace ~80% $\dot{V}O_2$ max [3:33 min/km], which was actually a personal best half-marathon for me. This final run was a personal psychological and physiological test inspired by the legendary Percy Cerruty (coach of Australian runner Herb Elliot), termed a "will power" session. As Percy himself said, "This session is the one that makes you who

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Jamie Stanley (center) pictured post marathon with Australian marathon record holder Robert de Castella (left) and father John Stanley (right), Australian national team member 1973 to 1985.

you are. Defines what you want to be and gives enlightenment to the individual of oneself. You only ever grow as a human being if you're outside your comfort zone." Percy's infamous "will power" session was done on a sand dune of about 30 to 60 m and the athletes were instructed to run up it as many times as their will allowed.

Throughout the entire preparation, I monitored my adaptation state with heart rate variability, which gives me useful insight, particularly when additional environmental stress loads (heat/hypoxia) were incorporated. In the end, I fell just short of my sub 02:30 debut marathon goal, completing it in 02:32:10 (at the time it was the 21st fastest marathon by an Australian in 2017) off an average weekly running distance of 79.1 km and working full time. To improve, I think that (slightly) more running volume might be required. However, this experiment proved to me that it's not all about "how can we do more?" and that asking "what is the minimum stimulus required for an optimal performance?" is equally an important question we can adopt to produce an optimal performance. In the words of Martin Buchheit, "The process of training is a journey of endless learning, but to learn one must have awareness of how the body adapts to a specific training stimulus."